Masoud Hemmatpour

mashemat@gmail.com
Date of Birth: May 29, 1983
Gender: Male
LinkedIn | GitHub | COM4AI

Residence:Oslo, Norway
Permanent Residency in Norway
Last update:01-08-2025
+47-46578143

PROFILE

Software and Systems Engineer specializing in **distributed systems, networking**, and **AI infrastructure**. 7+ years of experience designing high-performance data processing systems across research and industry environments. Experience collaborating with organizations such as **Cisco Systems**, **Nokia Bell Labs**, and **Dolphin Interconnect Solutions**. Strong background in **Linux systems**, **networking**, and **scalable computing platforms**.

PROFESSIONAL EXPERIENCE

- **Arctic University of Norway (UiT)** — Tromsø, Norway
  Research Engineer — *May. 2023 – Apr. 2025*
  - Designed and implemented an AI-driven framework that improved edge device energy efficiency by 4×, enabling more efficient distributed inference.
- **Simula Lab** — Oslo, Norway
  Research Engineer — *March. 2021 – Dec. 2023*
  - Developed in-network computing solutions to accelerate distributed data processing for applications including e-commerce analytics and smart grid monitoring.
- **Cisco Systems** — Paris, France
  Senior Software engineer — *Jan. 2019 – Jan. 2020*
  - Designed and implemented control and management plane solutions for cloud-native environments using C, Docker, Grafana, and Linux networking.

SELECTED PROJECTS

- **Horizon Europe MISO project**
  - Developed an intelligent framework that reduce energy consumption of end node up to 4 times.
  - Formulated work package ideas under our responsibility for project execution.
  - Participated in all steering, review, and technical meetings.
  - Delivered project deliverables and reports on time.
- **Linux Foundation hICN Project**
  - Designed and implemented models for software switch (i.e., VPP) to interface with network controllers, enabling remote telemetry and configuration capabilities.
  - Project demonstrated in Cisco Live.

PRACTICAL TRAINED COURSES

Supervised Machine Learning (Regression and Classification): build machine learning models in Python using popular machine learning libraries NumPy & scikit-learn

LPI (Linux Professional Institute): vendor-independent professional certification for Linux system administrators and programmers.

MCSA (Microsoft Certified System Administrator): a certification of the ability to configure or manage a windows server infrastructure or desktops and devices in an organization.

CCNA (Cisco Certified Network Associate): a certification of the ability to install, configure, operate, and troubleshoot medium-size routed and switched network.

TECHNICAL SKILLS

Programming Languages

C/C++ | Python | Bash

Networking

Ethernet | InfiniBand | RoCE

Operating Systems

Linux (Ubuntu,CentOS) | Windows

Data Storage

MySQL | Memcached | Redis

Distributed Systems

POSIX Threads | OpenMP | MPI

Machine Learning

PyTorch | STM32CubeAI | In-network AI

DevOps / Automation

CI/CD | GitHub Actions

Version Control

Git | Gerrit

Monitoring

YANG | sFlow | Grafana

Containerization

Docker | Kubernetes

EDUCATION

- **Politecnico di Torino** — Torino, Italy
  PhD in computer engineering — *Nov. 2015 – Sept. 2019*
  - **Thesis**
    - High Performance Computing using InfiniBand based clusters
  - **GPA: Graduated with distinction.**
- **Politecnico di Torino** — Torino, Italy
  MS in communications and computer networks engineering — *Oct. 2012 – Oct. 2015*
  - **Thesis**
    - Read-copy update synchronization technique improvement by hardware message passing
      Research project at École Politechnique Fédérale de Lausanne (EPFL)
  - **GPA: ≈ 3.93/4**

ACHIEVEMENTS

- Founder of COM4AI, a community focused on AI and communication systems.
- Pioneered memory storage over modern PCIe interconnect in HPC infrastructure.
- Developed a holistic fall prediction and detection system as part of the OPLON project.
- Best PhD dissertation award at Politecnico di Torino.
- Distinguished provincial mathematical prize 2000.

LANGUAGE SKILLS

- Norwegian, Intermediate, **B1**
- Italian, Upper Intermediate, **B2**
- English, **Fluent**
- Persian, **Native**

REFERENCES

Tore Heide Larsen
Chief Engineer at Simula, Oslo, Norway
torel@simula.no
+47-918 33 670

Tor Skie
Full Professor, University of Oslo (UiO), Norway
tskeie@ifi.uio.no
+47-97199620